



MedSecurance - IoMT Communication

Recommendation Tool (ICRT)

This toolset contains a comprehensive application designed for evaluating protocols, assessing risks, and managing and monitoring devices in real time.

Key Features:

- **Protocol Evaluation:** Analyze communication protocols to ensure compliance with industry standards and identify potential vulnerabilities.
 - Automated protocol analysis.
 - Detailed reporting on compliance and risks.
- **Risk Assessment**
 - Perform in-depth risk assessments for devices and network components.
 - Comprehensive risk scoring system.
 - Actionable insights for risk mitigation.
- **Device Management**
 - Seamlessly monitor and manage connected devices in your network.
 - Centralized device inventory.
 - Automated issue detection and alerts.
- **Real-Time Monitoring**
 - Stay updated with real-time data on device and network performance.
 - Live dashboards.
 - Event logging and detailed analytics.
 - Notifications for critical events or anomalies.

Main Components (containers):

- **Core components:** The core components comprise the backend, frontend, database and keycloak containers.
 - **backend:** The backend holds the main logic of the tool and connects the frontend with the database instance and the rest of the tool components such as the IDS, Monitor-Matrix, Email alerts, System Logs and Device Manager.
 - **frontend:** The frontend provides the User Interface (UI) to allow the user to interact with the tool. The main functionalities are:
 - Risk assessment
 - Device Manager
 - Administrator Configuration Panel
 - Activity Log Viewer
- **KeyCloack:** The keycloak container provide a centralize interface to handle the different user roles and user permissions for the different tool modules and components.
- **Database:** The database container utilizes an instance of a Postgres database that holds all necessary data generated by the different tools.
- **IDS:** The Intrusion Detection System (IDS) container allows the user to upload a network trace log file in a ".pcap" format to be analysed for any common network attacks. This module generates email alerts if an attack is detected.
- **Monitor-matrix:** The monitor-matrix container allows the user to monitor all device detected on a specific network and report the endpoints (domains/IPs) that the device is communicating with. This module generates alerts if the detected devices are not defined in the Device Manager by the network administrator (Unknown Device).
- **Monitor-matrix-lite:** The monitor-matrix-lite is identical to the Monitor-matrix container with the difference that it allows the user to manually activate the network scanning process for only IP addresses specified by the user. Note that no alerts are generated for Unknown Devices since the tool will only check IP addresses defined by the user.

Installation Manual

Overall, all modules of the tool can be deployed in any Operating System (OS) that Docker technologies can be deployed.

⚠ WARNING: Due to Docker technical limitations the Monitor-matrix container needs to be deployed on a Linux Based OS under the same network that the user wants to scan.

Step 1: Prerequisites (Installing Docker)

- For MAC OS follow this [instructions](#)
- For Windows OS follow this [instructions](#)
- For Linux OS follow this [instructions](#)
- For general instructions and Docker documentation use this [link](#)

Step 2: Fetching Tool source code

The tool source code can be downloaded from the following GitHub [repo](#). You can either download it as a ZIP file and extract it on your target machine or clone it using the git cli with the following command:

```
git clone https://github.com/MedSecurance/CommunicationRecommendation.git
```

① INFO: Note that the resulting folder name will be as follows depending on which method used to download the source code:

- Zip: CommunicationRecommendation-main
- Clone: CommunicationRecommendation

Step 3: Tool deployment

- First you need to enter the source code directory using the following command: “cd code-folder-name”, where code-folder-name must be replaced with the correct name from step 2 above.
- Next you need to edit the .env file using your favorite text editor (i.e., nano .env) and change the appropriate environmental variables based on the deployment type that you want to use as follow:
 - **POSTGRES_PASSWORD:** The database user that will be created during the database first startup (default = hackmeplease!)
 - **POSTGRES_USER:** The password of the database user that will be created during the first startup (default = medsec)

- **DB_CONNECTION_STRING:** The connection string of the database (default = "Host=postgres; Database=medsec; Username=medsec; Password=hackmeplease!")
- **EMAIL_HOST:** The host of your mail server (i.e. for gmail can be smtp.gmail.com) (default = "smtp.gmail.com")
- **EMAIL_HOST_PORT:** The port of the host mail server (default = "587")
- **EMAIL_SENDER_EMAIL:** The sender email (i.e. no-reply@something.com) (default = "MedSecurance_Example@example.com")
- **EMAIL_SENDER_PASSWORD:** The password of the sender email (default = "email_account_password_example")
- **EMAIL_SENDER_NAME:** The name of the sender to be shown on emails (default = "Med Securance")
- **REACT_APP_API_BASE_URL:** The URL of the Tool backend server. This is required for the frontend application to communicate with the backend (default = "<http://localhost:8080>")
- **ADMIN_EMAIL:** A valid email address for the network administrator. This email will receive notifications in case of intrusion detection events or unknown device detection (default = "network_admin_email@example.com")
- **UPDATE_DELAY_MINUTES:** Defines how often (in Minutes) the tool will request information for unknown devices on the network from the device manager module (default = "1")
- **EMAIL_DELAY_DAYS:** How frequently (in Days) the tool will send an "unknown device" email alert to the ADMIN_EMAIL (default = "1")
- **KEYCLOAK_BACKEND_URL:** The URL of the Keycloak server. This is required for the backend application to communicate with the Keycloak server (default = "<http://medsec-keycloak:8080>")
- **REACT_APP_KEYCLOAK_URL:** The URL of the Keycloak server. This is required for the frontend application to communicate with the Keycloak server (default = "<http://localhost:8081>")

 **WARNING:** Note that when a Gmail account is utilized you will need to configure it accordingly using the [Gmail API](#).

Step 4: Build and start Docker containers

To build and run the tool use the following command depending on the docker version that you are using:

- `docker compose up --build` **OR**
- `docker-compose up --build`

Step 5: Accessing Tool Modules

To access the different modules, use the following URLs:

- <http://localhost> to access the Risk Assessment, Device Manager, System Configuration Options and Log Files. Note that depending on the user role different Tool modules may not be visible.
 - Default username: user@example.com
 - Default password: password123
- <http://localhost:8081> to access the keycloak instance.
 - Default username: admin
 - Default password: admin
- http://localhost:33761/inspector_dashboard to access the Monitor-Matrix Module.
- <http://localhost:8501> to access the Intrusion Detection System (IDS)

❗ **INFO:** Note that the above deployment steps and configuration are recommended for local deployment for testing purposes. An example for production deployment is provided in the following sections of this manual.

Deploying individual Tool modules (containers)

To deploy individual components, you can comment out individual services that you do not need inside the docker-compose.yaml file:

- **Core modules:** To deploy only the core modules you need to keep the following lines corresponding to their docker containers and data volumes:
 - postgres: lines 2 - 17
 - backend: lines 19 - 39
 - frontend: lines 41 - 52
 - keycloak: lines 88 – 102
 - data volumes: lines 114 - 117
- **IDS module:** To deploy only the ids container* you need keep only the following lines:
 - ids: lines: 77 - 86
 - data volumes: lines 114 – 117

* Note that email alerts will not work if the Core modules are not deployed
- **Monitor-Matrix module:** To deploy only the Monitor-Matrix container** you need to keep only the following lines:
 - monitor-matrix: lines 57 – 75
 - data volumes: lines 114 – 117

** Note that email alerts and unknow device detection will not work if the Core modules are not deployed

❶ **INFO:** To substitute **Monitor-matrix** with **Monitor-matrix-lite** see inline comments in the main “docker-compose.yml” file. You can also deploy it as a stand-alone service using the “docker-compose.yml” located inside the Monitor-matrix-lite folder.

Production deployment (Linux OS)

In this scenario, we assume that the Core modules and the IDS module are deployed on a different machine (centralized server) with the IP address 192.168.1.1 and the Monitor-matrix module is either on the same or a different machine with a different IP address, i.e., 192.168.1.2. Note that we can also use URLs instead of an IP address.

In this scenario we need to alter the following files during the installation process in Step 3 as follows:

- .env file
 - REACT_APP_API_BASE_URL=“<http://192.168.1.1:8080>”
- docker-compose.yml (monitor-matrix service):
 - Uncomment lines 59: network_mode: “host”
 - Change line 69 to BACKEND_URL: “<http://192.168.1.1:8080>”
 - Comment the lines 74 and 75 as follows:
 - # ports
 - # - “33761:33761”

💡 **TIP:** See “Deploying individual Tool modules (containers)” section on how to deploy Monitor-matrix module and Core modules separately.

❶ **INFO:** For **Monitor-matrix-lite production deployment under Linux OS** see inline comments in the main “docker-compose.yml” file located inside the Monitor-matrix-lite folder.

After the above changes you can continue with the remaining installation steps (Step 4 and 5) as normal to deploy the corresponding modules to the appropriate physical machines. Note that for step 5 you will need to replace the “localhost” with the IP address or URL address of the corresponding module i.e., for Monitor-matrix should be 192.168.1.2 and for the rest modules 192.168.1.1.

The above changes will allow the Monitor Matrix module to communicate with the remote deployment of the Core components. Email alerts and Unknown device

detection functionality will work as expected and the Monitor-matrix component will be able to scan the entire network IP range (192.168.1.1 – 192.168.1.255).

 **WARNING:** To deploy Monitor-matrix tool on an entirely different network, a proper URL needs to be configured for the Core modules to allow Monitor-matrix to communicate with it over the Internet.

User Manual

Core modules – Risk Assessment Wizard

1. Open your favorite web browser and navigate to the following url: <http://server-ip-address> (where “server-ip-address” should be replaced with the actual deployment IP address or domain name).
2. The following windows will appear:

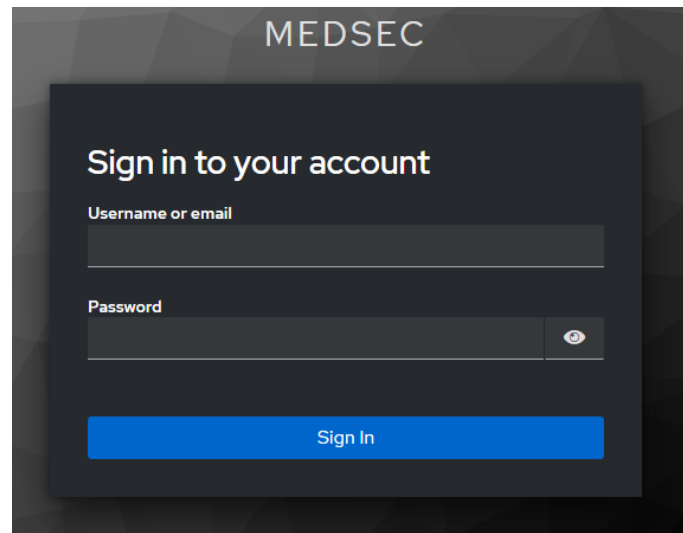


Figure 1: Initial login window of the Core Modules

3. The default credentials for the administrator user are:
 - a. Username: user@example.com
 - b. Password: password123
4. After a successful login the following window will appear:

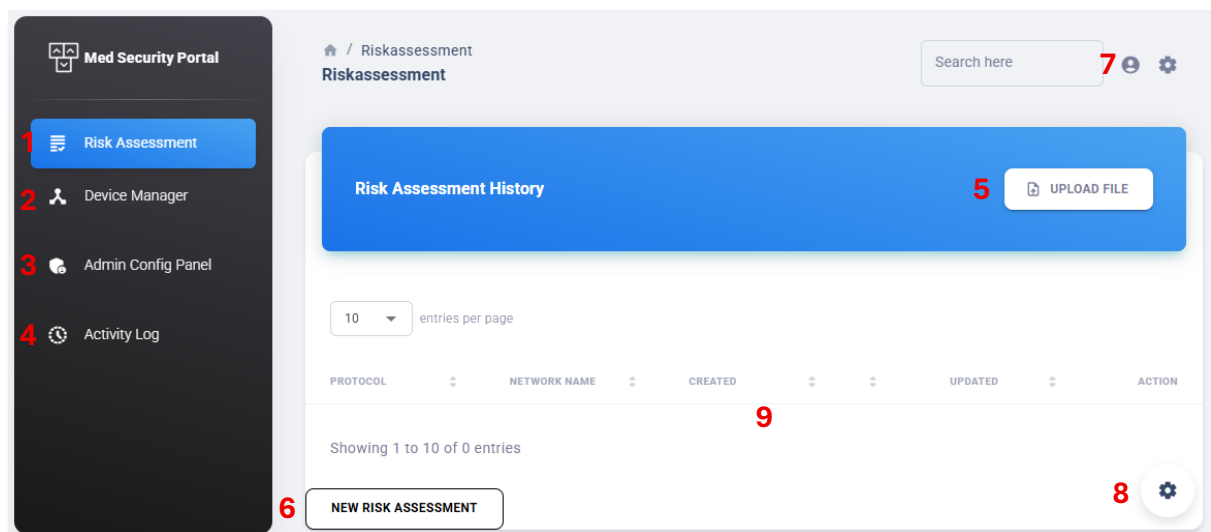


Figure 2: Main user interface for the Core modules

5. The Main user interface (UI) for the Core modules provides the following functionality as depicted with the red numbers in Figure 2:
 1. **Risk Assessment:** The risk assessment option allows the user to start a new risk assessment process. This is the default view when accessing the Core modules UI.
 2. **Device Manager:** The device manager option allows the user to access the device management functionalities including listing, adding, deleting and updating the different devices attached to the different networks defined in the risk assessment processes.
 3. **Admin Config Panel:** The admin config panel allows the user to configure different constant values related to the risk assessment process, such as mean time to repair, mean down time and expected lifetime of networks.
 4. **Activity Log:** The activity log allows the user to view all the users' actions in the system, including adding new devices, executing new risk assessments, updating devices, etc.
 5. **UPLOAD FILE:** This functionality is part of the risk assessment process and allows the user to upload a risk assessment process in JSON format. Note that a risk assessment process can be downloaded in a JSON format upon initial creation (See below for more detail).
 6. **NEW RISK ASSESSMENT:** The new risk assessment option allows the user to initiate a new risk assessment process.
 7. **User Account:** The user account option allows the user to view the information of the current user and provides the Logout functionality.
 8. **User Interface Option:** The user interface option allows the user to customize the UI, i.e., change the theme from Light to Dark mode.
 9. **Risk Assessments list view:** The risk assessment list view includes a list of all risk assessments currently available in the system. The user can use it to edit, update and delete existing risk assessment processes.
6. When the user selects the “NEW RISK ASSESSMENT” option (Figure 2 – Option 6), a new risk assessment process is initialized, and the following window will appear (See Figure 3). Currently the risk assessment process supports four protocols Wi-Fi, Bluetooth, GSM and LoRa WAN.

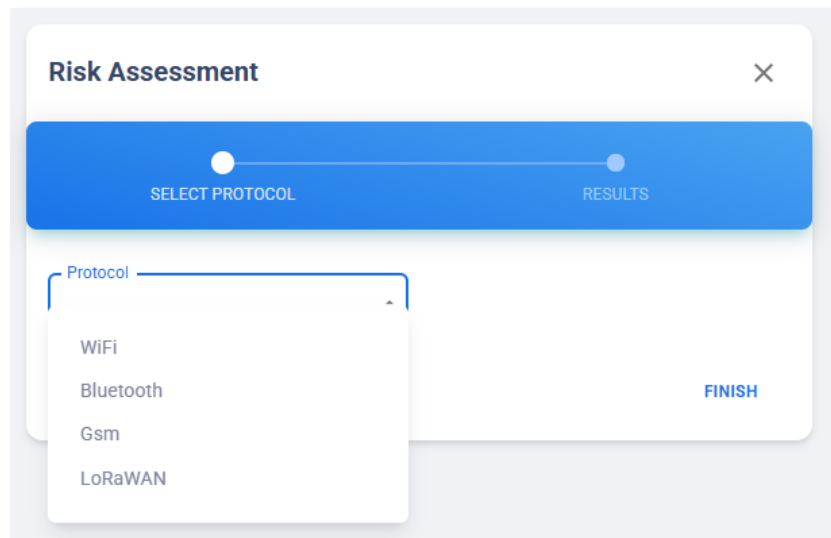


Figure 3: New Risk Assessment Process-Protocol selection options

7. Depending on the selected protocol a new window will appear. Figure 4 shows the risk assessment wizard for the Wi-Fi protocol.

Risk Assessment

SELECT PROTOCOL RESULTS

Protocol
WiFi

1 Description 2 UseCase

3 Network Name 4 ☐ Firewall Enabled

5 ☐ Log Monitor Enabled 6 ☐ Redundancy Measures

7 ☐ Intrusion Detection System 8 ☐ Firmware Integrity Check

9 Ip Range 10 ☐ Already Implemented

11 Backbone Network Speed In Mbps 12 ISP Connection Speed In Mbps

Network Details

13 ADD ACCESS POINTS 14 Other Connected Devices
0

15 Topology Type
SingleAP 16 Lifetime in Years

17 Placement 18 ADD NETWORK FAILURES

19 UPLOAD TVRA INPUT

BACK 20 FINISH

Figure 4: Risk Assessment wizard for the Wi-Fi protocol

8. The risk assessment wizard for the Wi-Fi protocol includes 20 steps as follows depicted with red numbers in Figure 4:
1. **Description:** The Wi-Fi network description, i.e., Intensive case Wi-Fi network.

2. **Use Case:** A short description of the Wi-Fi network use case, i.e., Connects all Wi-Fi equipment in the intensive care units, including sensors, actuator and data collectors.
3. **Network Name:** A user defined network name to help the user identify the physical network in the system, i.e., Intensive-Care-Wi-Fi-Net.

 **WARNING:** Note that the “Network Name” must be the same for the devices attached to the network when defined in the Device Manager module.

4. **Firewall Enable:** The user shall select this option when the network is protected by a firewall.
5. **Log Monitoring Enabled:** The user shall select this option when the network has log monitoring functionality enabled.
6. **Redundancy Measures:** The user shall select this option when the network has redundancy measures enabled.
7. **Intrusion Detection System:** The user shall select this option when the network has an intrusion detection system enabled.
8. **Firmware Integrity Check:** The user shall select this option when the network admin has a firmware integrity check policies in place for all components on the network.
9. **IP Range:** The IP Range option must be in a [CIDR format](#). i.e., 192.168.1.0/24
10. **Already Implemented:** This option shall be selected if the network is not actually implemented but rather the user is exploring the possibility of deploying such a network. This option allows the system to be more “aggressive” with respect to the final recommendations and suggestions since we are assessing a hypothetical network and no purchase took place up to this point.
11. **Backbone Network Speed in Mbps:** This value should be the speed in Mbps of the wired network that connect the Wi-Fi accesspoints to the rest of the internal network.
12. **ISP Connection Speed in Mbps:** This value should be the speed in Mbps of the Internet Service Provider that connects the internal network to the Internet.
13. **Add Access Points:** This button allows the user to add access points (AP) to the network (see below for more details).
14. **Other Connected Devices:** This value should be more than zero if the network is expected to have additional connected device that are not already defined in the Device Manager module.
15. **Topology Type:** The user had to choose one of the following options depending on the actual network deployment:
 - a. **SingleAp:** For a network with just one AP
 - b. **Star:** For a network with more than one AP configured in a Star topology
 - c. **Mesh:** For a network with more than one AP configured in a Mesh topology

16. **Lifetime in Years:** The expected lifetime in years of the network, i.e., 5 years.
17. **Placement:** The user had to choose one of the following options depending on the actual network physical or logical placement:
- Local:** When the network is deployed locally, within the organization premises.
 - Cloud:** When the network is deployed on a cloud service.
 - Hybrid:** When the network is deployed on a both local and cloud premises.
18. **Add Network Failures:** This button allows the user to add different type of network failures (see below for more details).
19. **Upload TVRA Input:** This button allows the user to upload the output of TVRA (if available) to improve the overall recommendations and suggestions of the risk assessment process.
- 💡 **TIP:** The TVRA input is optional and can be uploaded at any time by updating the risk assessment wizard when TVRA output is available.
20. **Finish:** The user shall click the finish button when all of the above inputs are provided. In case of missing or wrong values appropriate messages will appear to help the user to resolve any issues. For the final recommendations and suggestions see below.
9. Following the description in 8.13 (Add Access Point), when the user click the button the following window will appear.

The screenshot shows a web-based wizard titled "Add Access Points". It contains several input fields and a button, each marked with a red number from 1 to 11. The fields are: "Access Point Name" (1), "ssid" (2), "Hidden ssid" (3, with a checkbox), "Manufacturer" (4), "Model" (5), "Firmware updated year" (6), "Antenna type" (7, with a dropdown arrow), "Physical location" (8, with a dropdown arrow), and an "ADD STANDARD" button (9). At the bottom right, there are "CANCEL" (10) and "SAVE" (11) buttons.

Figure 5: Wi-Fi protocol – Adding an Access Point wizard

10. The Access Point wizard is an 11 steps process as depicted in Figure 5 with red colored numbers:

1. **Access Point Name:** A user defined AP name, i.e., AP1.
 2. **ssid:** The actual service set identifier ([SSID](#)) of the AP, i.e., ap1_2.4Ghz
 3. **Hidden ssid:** The user shall select this option when the AP ssid is hidden, that is, configuring the AP to not broadcast its ssid publicly.
 4. **Manufacturer:** The AP manufacturer name, i.e., Cisco, TP-Link, Netgear, etc.
 5. **Model:** The actual model of the AP, i.e., Netgear WAX630, Cisco Meraki MR57, etc.
 6. **Firmware updated year:** The year that the AP firmware was last updated, i.e., 2023.
 7. **Antena Type:** The user should choose one of the following options, omnidirectional, directional, batch, yagi and panel.
 8. **Physical location:** The user should choose one of the following options:
 - a. **secure place:** When the AP is in a secure place (i.e., in a server room)
 - b. **open space:** When the AP is in an open space accessible by the public, i.e., within the patient's waiting area.
 - c. **private place:** When the AP is located within a private place, i.e., within a doctor's office.
 9. **Add Standard:** This button allows the user to define the standards that the AP can support (see below for more details).
 10. **CANCEL:** The user can click this button to cancel the add AP wizard.
 11. **SAVE:** The user can click this button to save the AP wizard. In case of missing or wrong values appropriate messages will appear to help the user to resolve any issues.
11. Following the description in 10.9 (Add Standard) the following window will appear. The user can choose one of the following options, 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac and 802.11ax.

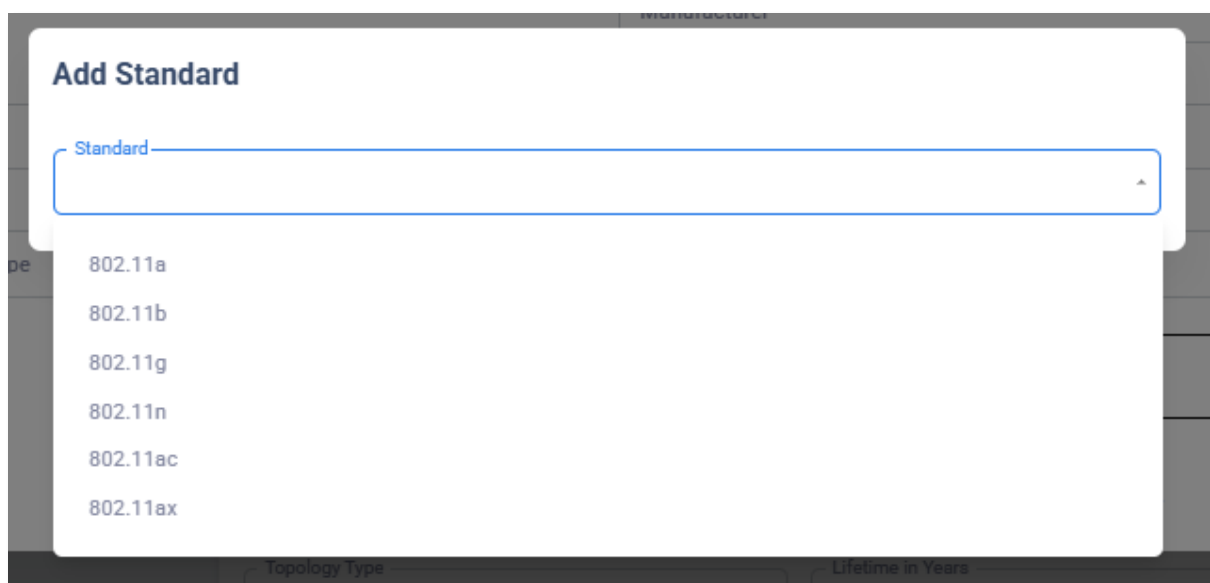


Figure 6: Wi-Fi protocol – Adding an AP wizard – Add Standard options

12. Depending on the user selection in step 11 (see Figure 6) the user can then select the appropriate *Operational Frequencies* and *Encryption Utilized* options as depicted in Figure 7.

Add Standard

Standard
802.11ax

Operation Frequencies

- 2_4 GHz
- 5 GHz
- 6 GHz
- 2_4 GHz AND 5GHz
- 2_4 GHz AND 6GHz
- 5 GHz AND 6GHz
- 2_4 GHz AND 5GHz AND 6GHz

Encryption Utilized
WPA3

CANCEL

Figure 7: Wi-Fi protocol – Adding an AP wizard – Add Standard additional options

13. The user can select more than one standard per AP with the minimum of at least one standard. In cases where the AP can support more than one standard, then the user needs to select which one is utilized as depicted in Figure 8 (highlighted by a red box) by selecting the corresponding check box next to the utilized standard.

Add Access Points

Access Point Name: Access_point_1

ssid: access_point_2.4

☐ Hidden ssid

Manufacturer: Zyxel

Model: model1

Firmware updated year: 2024

Antenna type: omnidirectional

Physical location: open space

☐ 802_11a x
☐ 802_11b x
☐ 802_11g x
☐ 802_11n x
☐ 802_11ac x
☒ 802_11ax x

ADD STANDARD

CANCEL SAVE

Figure 8: Wi-Fi protocol – Adding an AP wizard – Selecting utilized standard

14. Finally, the user can either select SAVE to save the AP or CANCEL to return to the risk assessment wizard without inserting an AP.
15. Following the description in 8.18 (Add Network Failures) by clicking on the button the following window will appear:

Add Network Failures

1 Cause Of Failure: Hardware Issue

2 Downtime In Minutes: 30

3 Time To Repair In Minutes: 30

4 Failure Handling: Replace faulty component

5 CANCEL SAVE

Figure 9: Wi-Fi protocol – Add Network Failures UI form

16. To add a network failure is a 5-step process as depicted in Figure 9:

1. **Cause of Failure:** The user can choose one of the following options:
 - a. **Hardware Issue:** When a hardware failure occurs, i.e., faulty AP.
 - b. **Interference:** When interferences affect the APs connection, i.e., areas saturated by multiple APs.
 - c. **Overload:** When the Wi-Fi network is overloaded due to high data bandwidth, i.e., too much traffic generated by the connected devices on the network.
 - d. **Software Firmware:** Issues related to software or firmware of the devices or APs, i.e., AP firmware issue or device software issues.
 - e. **Power:** Issues related to power failures, i.e., frequent power outages.
 - f. **Environment:** Issues related to the environment, i.e., high temperature, storms, hurricanes, etc.
 - g. **Configuration Error:** Issues related to configuration errors, i.e., IP conflicts, DHCP issues, routing issues, etc.
 2. **Downtime in Minutes:** The time in minutes that the network was down due to the failure.
 3. **Time to Repair in Minutes:** The time in minutes that it took to fix the failure.
 4. **Failure Handling:** A short description of how the failure was handled.
17. Following the description in 8.20 (FINISH – Risk Assessment wizard) the following window will appear (see Figure 10).
18. The risk assessment wizards' results UI provides the following functionality as depicted with red numbers in Figure 10:
1. **Mitigations:** This section presents the process mitigations suggestions.
 2. **Safe configurations:** This section presents the process safe configuration suggestions.
 3. **Protocol replacement:** This section presents protocol replacement suggestions.
 4. **Vulnerabilities:** This section presents any vulnerabilities that are related to the devices attached to the network as analyzed by the TVRA tool. Note that this section will be populated only if a TVRA report is provided.
 5. **BACK:** The user can select this button to go back to the risk assessment wizard to update any of the provided information.
 6. **DOWNLOAD ANSWERS:** This button allows the user to download the risk assessment locally in a JSON format and reupload it to the system as mentioned in step 5.5 (see Figure 2).
 7. **DOWNLOAD PDF:** This button allows the user to download the risk assessment locally in a human readable format (PDF file).
 8. **X:** This button allows the user to remove/ignore a recommendation from appearing to the human readable report (PDF). Note that it will still be available in the JSON format report.

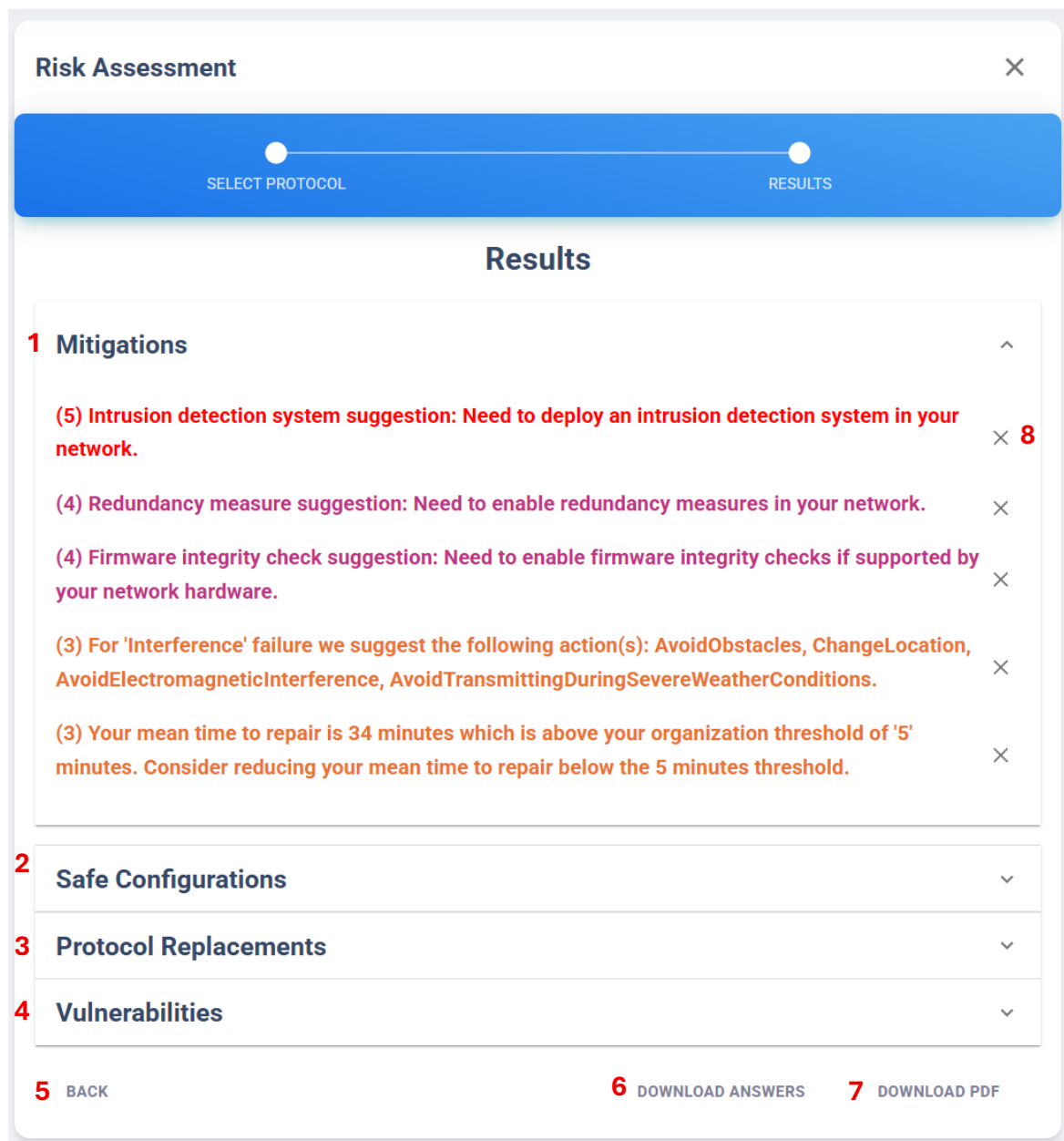


Figure 10: Wi-Fi protocol risk assessment wizard results

19. Finally, Figure 11 shows the Risk Assessment list view after a risk assessment has been successfully completed. The user can interact with a specific risk assessment as depicted by the two red numbers in Figure 11 as follow:

1. Edit/update the risk assessment wizard by clicking on the number 1 icon.
2. Delete the risk assessment wizard by clicking on the number 2 icon.

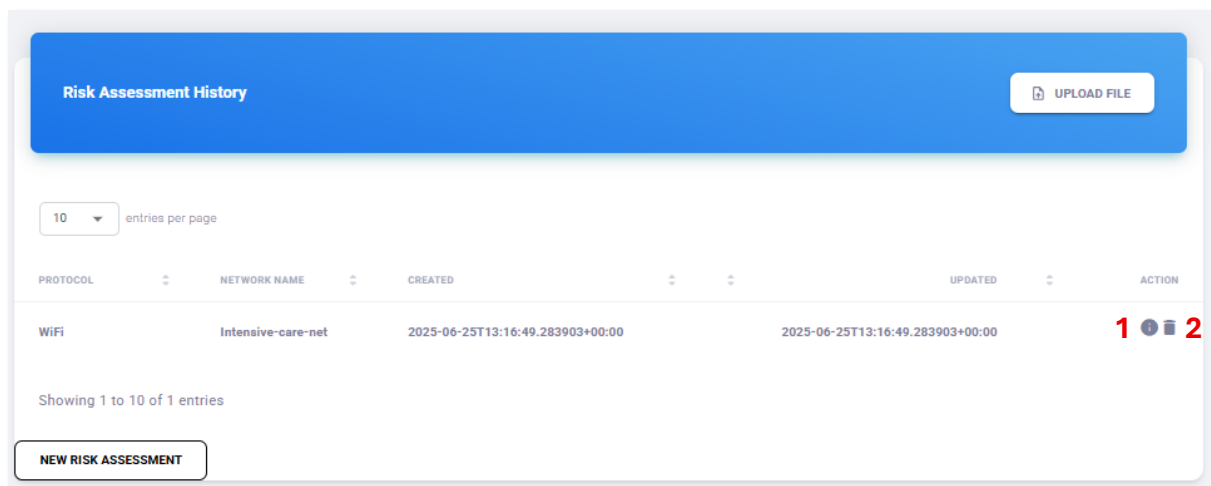


Figure 11: Risk assessment list view interface after a risk assessment has been completed

Core modules – Device Manager

1. To access the “Device Manager” UI the user should click on the corresponding button (highlighted with a red box in Figure 12).

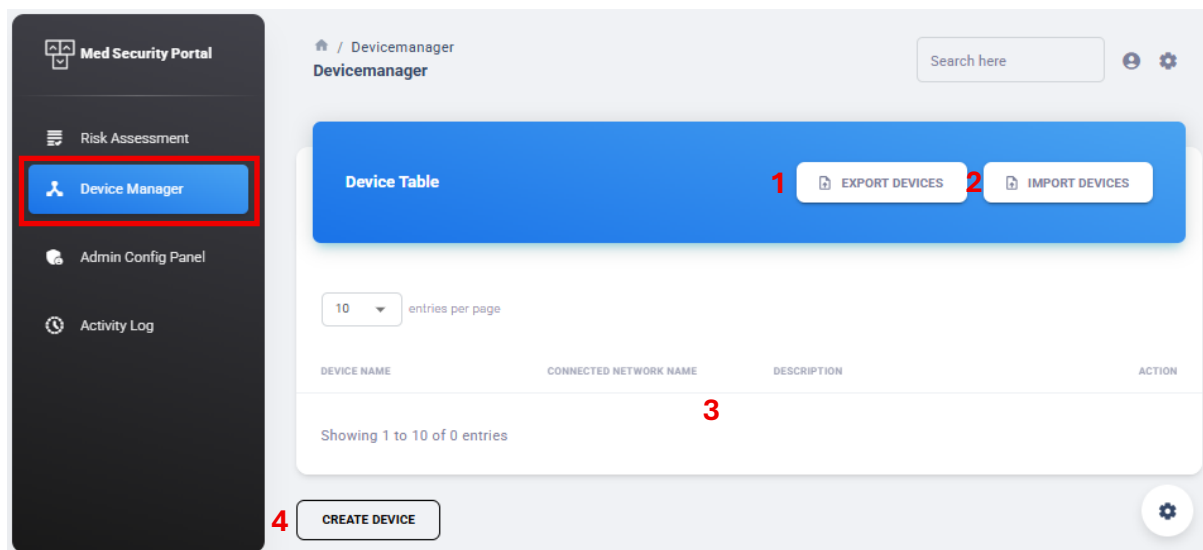


Figure 12: Device Manager Module UI

2. The Device Manager Module UI provides the following functionality:
 1. **EXPORT DEVICES:** This button allows the user to export all devices available in the Devices list view as a single JSON Object.
 2. **IMPORT DEVICES:** This button allows the user to import devices from a JSON Object (generated by the EXPORT DEVICES functionality).
 3. **Devices list view:** The view shows all devices already defined in the Device Manager. See below for more details related to the provided functionality.

4. **CREATE DEVICE:** This button allows the user to create a new Device. See below for more details related to the device's creation wizard.
3. To starts the “Create device wizard” the user should click on the corresponding button described in 2.4 and depicted in Figure 12 (Option number 4). The following window will appear:

The screenshot shows a 'Create New Device' form with the following elements and step numbers:

- 1** Name (text input)
- 2** Serial Number (text input)
- 3** Type (dropdown menu)
- 4** Manufacturer (text input)
- 5** Description (text input)
- 6** Manufacturing Date (calendar icon, pre-filled with 07/03/2025)
- 7** Network Name (text input)
- 8** Doctor ID (text input)
- 9** Location (dropdown menu)
- 10** ☐ Validated
- 11** ☐ Standard Compliance
- 12** Communication Protocols section:
 - ☐ WiFi
 - ☐ Bluetooth
 - ☐ GSM
 - ☐ LoraWan
- 13** Communication Protocol (dropdown menu)
- 14** CANCEL and CREATE buttons at the bottom right.

Figure 13: Device Manager – Create Device wizard

4. The Create Device wizard is a 14-step process as depicted with red numbers in Figure 13:
1. **Name:** A user defined device name
 2. **Serial Number:** The serial number of the device
 3. **Type:** A dropdown with three options:

- a) **Sensor:** Select this option if the device is a sensor, i.e., blood pressure meter.
 - b) **Actuator:** Select this option if the device is an actuator, i.e., an insulin pump.
 - c) **Collector:** Select this option if the device is collecting data from other devices on the network, i.e., smart phone or tablet.
4. **Manufacturer:** The name of the device manufacturer.
 5. **Description:** A short description of the device.
 6. **Manufacturing Date:** The device manufacturing date.
 7. **Network Name:** The network name that the device is connected to.

 **WARNING:** Note that the “Network Name” must be the same with the network name is the Risk Assessment wizard (see Risk Assessment Wizard point 8.3).

8. **Doctor ID:** The doctor’s identifier as defined by the user that is utilizing the devices. This value can be set to “None” if no doctor is assigned to the device.
9. **Location:** The user should choose one of the following options:
 - a) **secure place:** When the device is in a secure place (i.e., in a server room)
 - b) **open space:** When the device is in an open space accessible by the public, i.e., within the patient’s room.
 - c) **private place:** When the device is located within a private place, i.e., within an intensive care unit.
10. **Validated:** The user should select this option if the device is valid, i.e., the user already confirms that the device is manufactured by a credible manufacturer by scanning the appropriate barcode to verify the device authenticity. This option can be updated at any time after the initial creation of the device.
11. **Standard Compliance:** The user should select this option if the device complies with any standards. This is a manual process that needs to be done by the user. This option can be updated at any time after the initial creation of the device.
12. **Communication Protocols (Checkboxes):** The user can select one or more communication protocols that the device can support. Each checkbox will include a new section at the bottom of the form to allow the user to provide more details related to the device capabilities for the corresponding protocol.
13. **Communication Protocol (Dropdown):** The user can select the currently utilized communication protocol which can be one of those already selected in the checkboxes in the previous step.
14. **CANCEL & SAVE:** Finally, the user can either select the SAVE button to save the device or the CANCEL button to abort the new device creation.

5. Finally, Figure 14 shows the Device Manager list view after the successful creation of a new device. The user can interact with a specific device as depicted by the two red numbers in Figure 14 as follows:
 1. Edit/update the risk assessment wizard by clicking on the number 1 icon.
 2. Delete the risk assessment wizard by clicking on the number 2 icon.

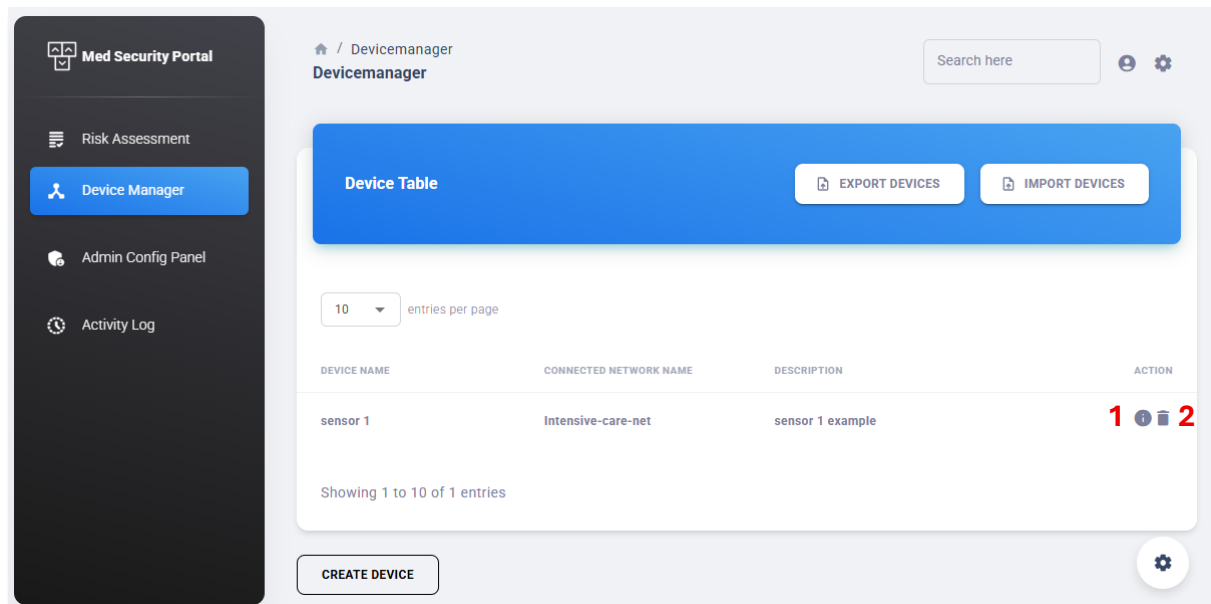


Figure 14: The Device Manager list view after a new device wizard has been completed

Core modules – Admin Config Panel

1. To access the “Admin Config Panel” UI the user should click on the corresponding button (highlighted with a red box in Figure 15).
2. The Admin Config Panel UI provides the following functionality:
 - a. The highlighted numbers 1, 2, 3 and 4 group together the constant configuration values for the Wi-Fi, Bluetooth, GSM and LoRa WAN communication protocols, respectively.
 - b. Each communication protocol group includes the following 3 constants that the user can configure as follows:
 - i. **LifetimeInYears:** This constant defines the maximum lifetime of a network. If a network is older than this constant a recommendation will appear when running a risk assessment process on a network.
 - ii. **MeanDowntimeInMinutes:** This constant defines the maximum mean downtime of a network. If a network failure takes more time to recover from a failure a recommendation will appear when running a risk assessment process on a network.

- iii. **MeantimeToRepair:** This constant defines the maximum mean time to repair a network failure. If a network failure takes more time to be repaired a recommendation will appear when running a risk assessment process on a network.
- c. Finally, by using the icons next to the red number 5 in Figure 15, the user can change the corresponding value of each one of the options. For different Communications protocols, the user first needs to click on the corresponding group and then the constant values will be visible.

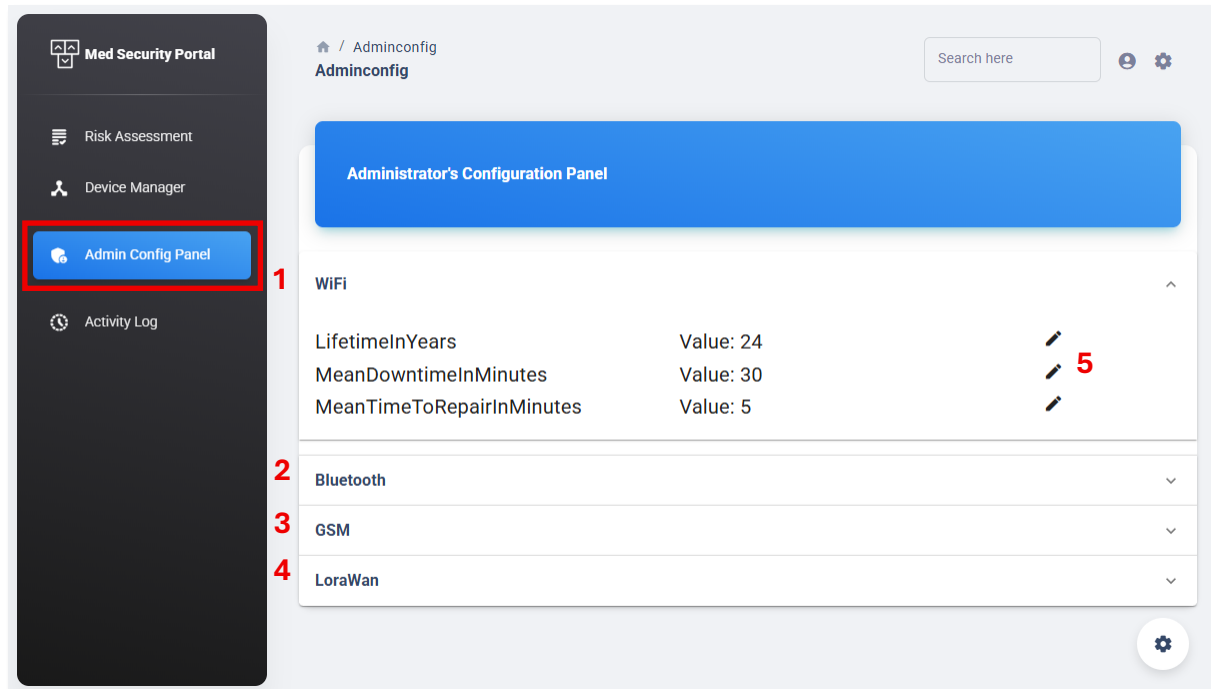


Figure 15: The Admin Config Panel UI

Core modules – Activity Log

1. To access the “Activity Log” UI the user should click on the corresponding button (highlighted with a red box in Figure 16).
2. The Activity Log list view (left side of Figure 16) provides the following information for each user’s action in the system:
 - a. **User ID:** The internal unique user ID
 - b. **USER FULL NAME:** The full name of the user
 - c. **CATEGORY:** The module that the user action was observed
 - d. **ACTION:** The user action that was observed (Add, Update, Delete and Execute)
 - e. **DETAILS:** The details of the user action
 - f. **CREATEDAT:** The timestamp of the recorded user action

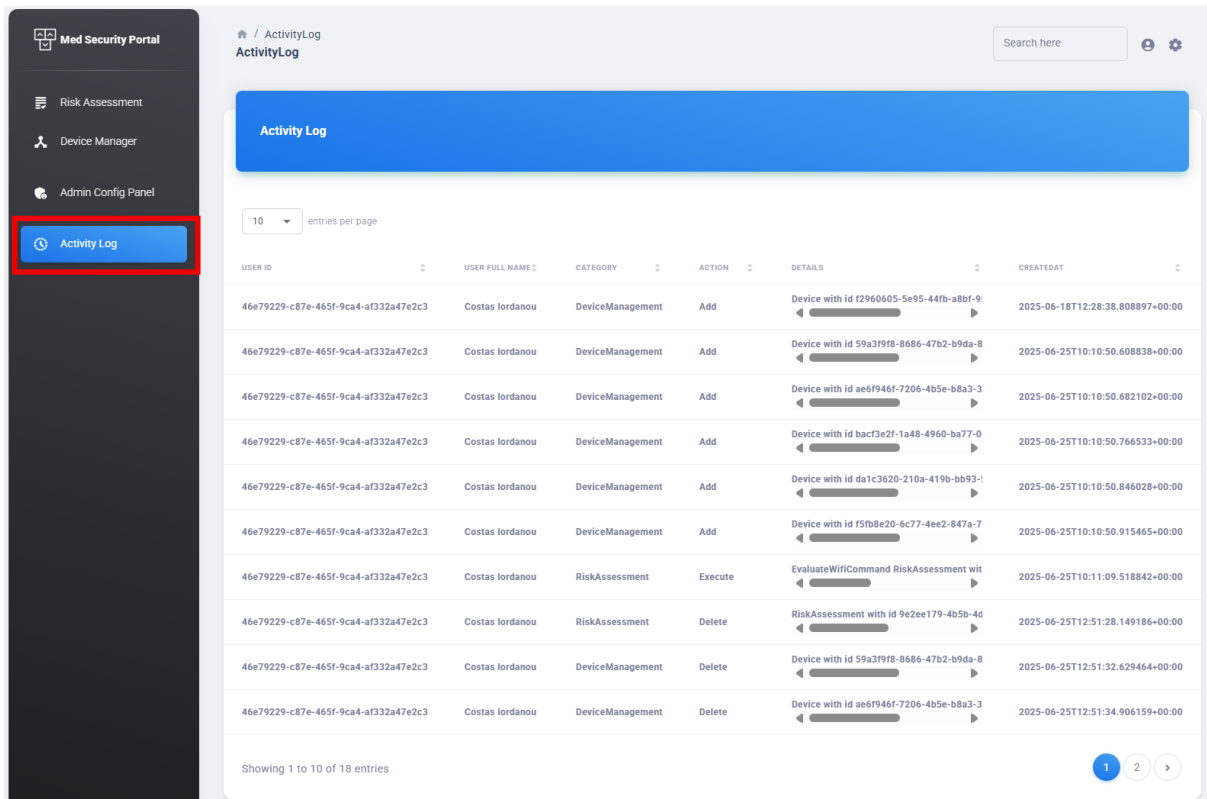


Figure 16: The Activity Log UI

Intrusion Detection System (IDS) Module

1. To access the IDS system, open your favorite web browser and navigate to the following url: <http://server-ip-address:8501> (where “server-ip-address:port” should be replaced with the actual deployment IP address or domain name).
2. The following windows will appear (see Figure 17).

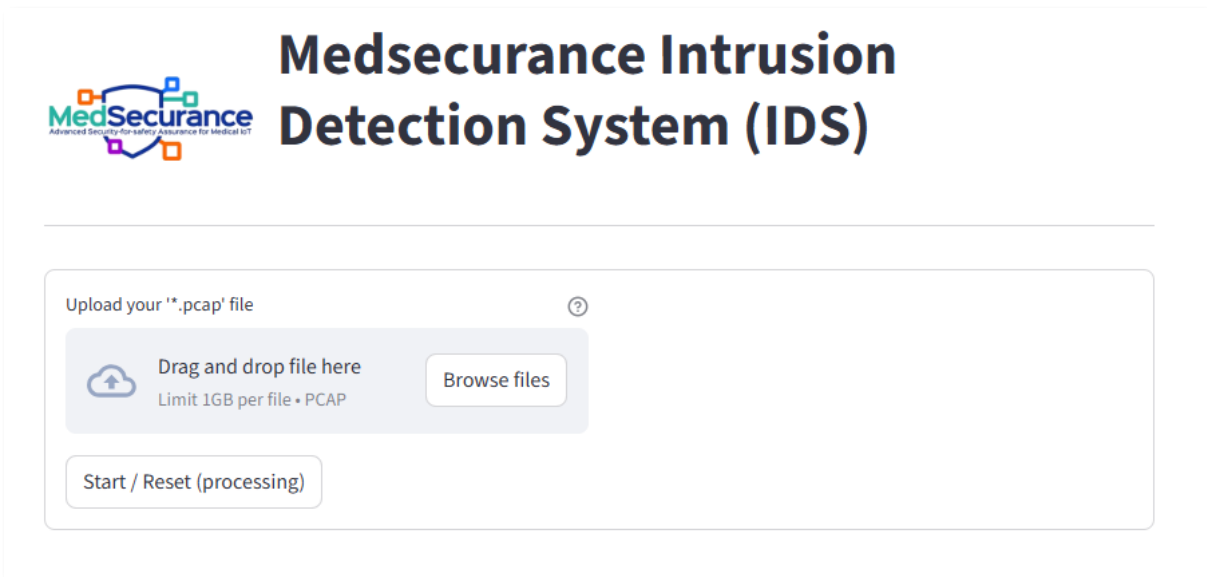


Figure 17: The Intrusion Detection System UI

3. Overall, utilizing the tool is a 3-step process as depicted in Figure 18:
 1. First, the user needs to click on the “Browse files” button. A new file selector will appear to allow the user to select any network trace file (.pcap). Note that the maximum file size is 1GB.
 2. Upon successful selection the selected file will be uploaded to the system and appear below the “Browse files” button area.
 3. Finally, the user can click on the “Start / Reset (processing)” button to start the file analysis.

MedSecurance Intrusion Detection System (IDS)

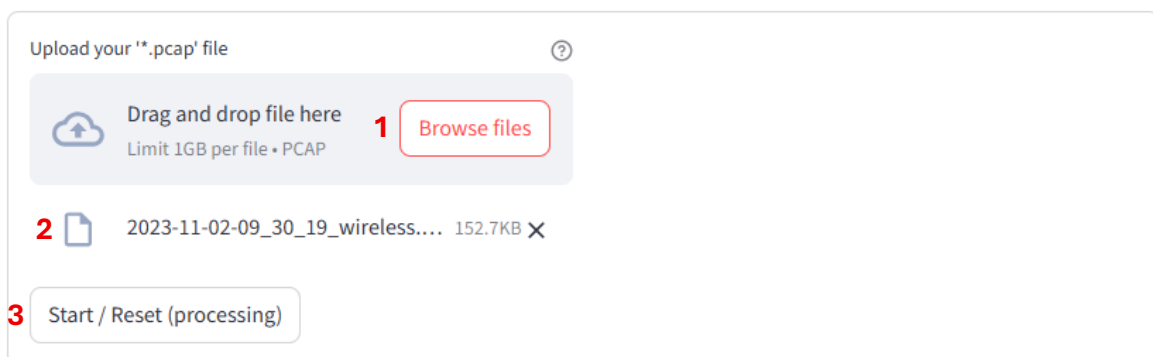


Figure 18: IDS – Uploading a network trace file

4. Figure 19 shows the final results of the uploaded file analysis as follow:

- The results list view provides the following information:
 - Source MAC address:** The MAC address of the source device
 - Destination MAC address:** The MAC address of the destination device.
 - Attack detected:** The checkbox is checked when an attack is detected on the specific flow
 - Attack probability:** Is either 1 if an attack is detected, otherwise is 0
 - Attack Type:** If the system detects an attack on the specific flow then this column will include the results related to the type of the attack, otherwise it will be empty.
- The user can also download, search or zoom in the results list view by clicking on any of the buttons next to the red number 2 in Figure 19, respectively.
- Finally, by double clicking on any row "Attack Type" column, the user can also see the probability of each attack type.
- When an attack is detected an email alert is send to the network administrator.



Medsecurity Intrusion Detection System (IDS)

Upload your '*.pcap' file ?

Drag and drop file here

Limit 1GB per file • PCAP

Browse files

Start / Reset (processing)

Input file: 2023-11-02-09_30_19_wireless.pcap - Size: 0.15 MB

1	Source MAC address	Destination MAC ad	Attack detected	Attack Probability	Attack Type
0	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	["attack_type": "Web", "probs": {"Benign": 0, "BruteForce": 0, "DDoS": 0, "DoS": 0, "Mirai": 0, "Recon": 0, "Spoofing": 0, "Web": 1}]
1	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	3 <pre> { "attack_type": "Web" "probs": { "Benign": 0 "BruteForce": 0 "DDoS": 0 "DoS": 0 "Mirai": 0 "Recon": 0 "Spoofing": 0 "Web": 1 } } </pre>
2	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	
3	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	
4	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	
5	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	
6	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	
7	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	
8	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	
9	34:e1:2d:ed:a7:29	01:00:5e:00:00:01	☑	1	

Attacks detected.

4 Filename: 2023-11-02-09_30_19_wireless.pcap

Send email to cos.ior@cytanet.com.cy

Figure 19: IDS – Inspecting the results

Monitor-Matrix Module

1. To access the Monitor-Matrix module, open your favorite web browser and navigate to the following url: `http://server-ip-address: :33761/inspector_dashboard/` (where “server-ip-address:port” should be replaced with the actual deployment IP address or domain name).
2. The following windows will appear (see Figure 20). To start the Monitor-Matrix process the user needs to accept the risks (Red button highlighted with number 1) explained in the initial page.

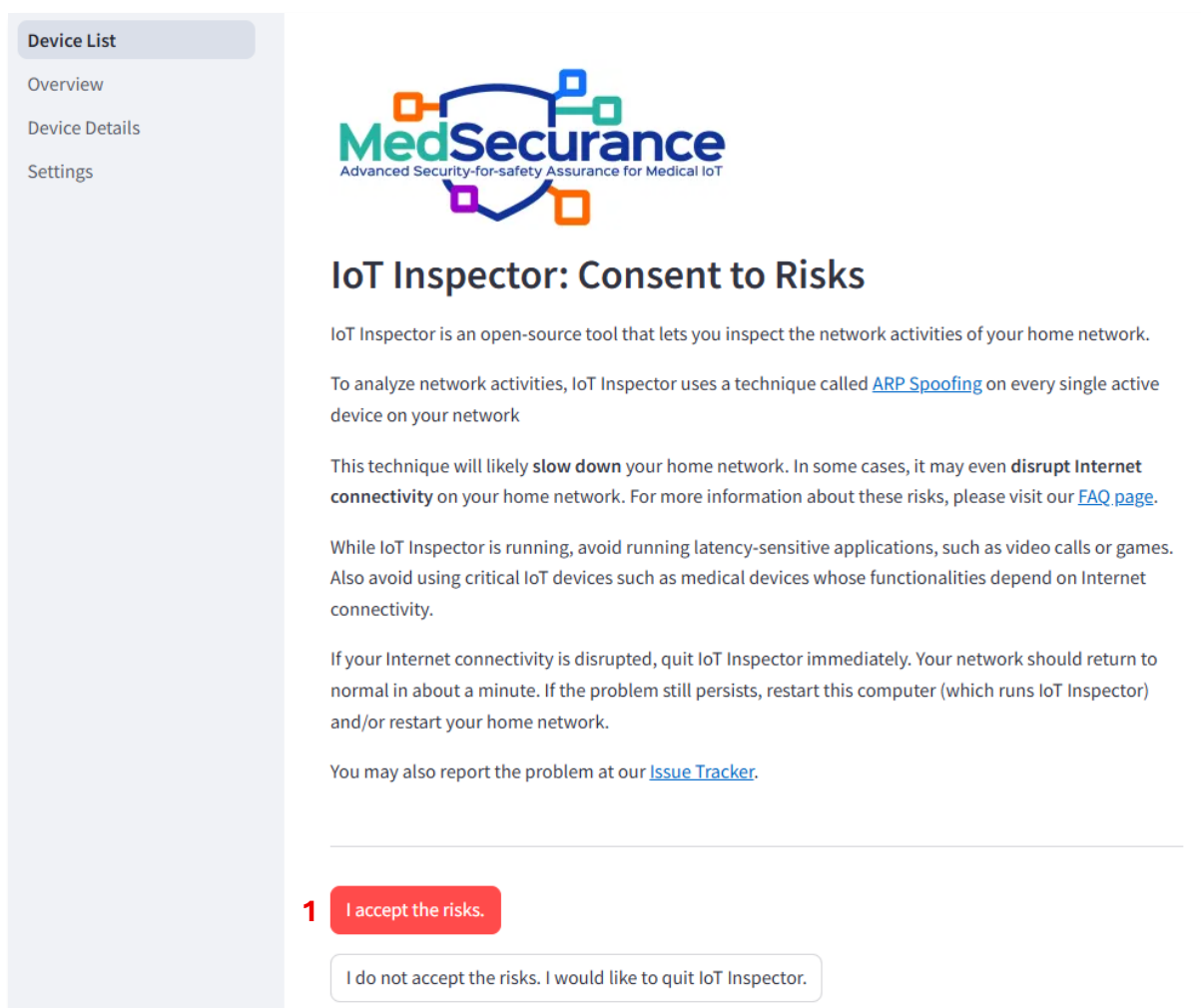


Figure 20: Monitor-Matrix Consent to Risks UI

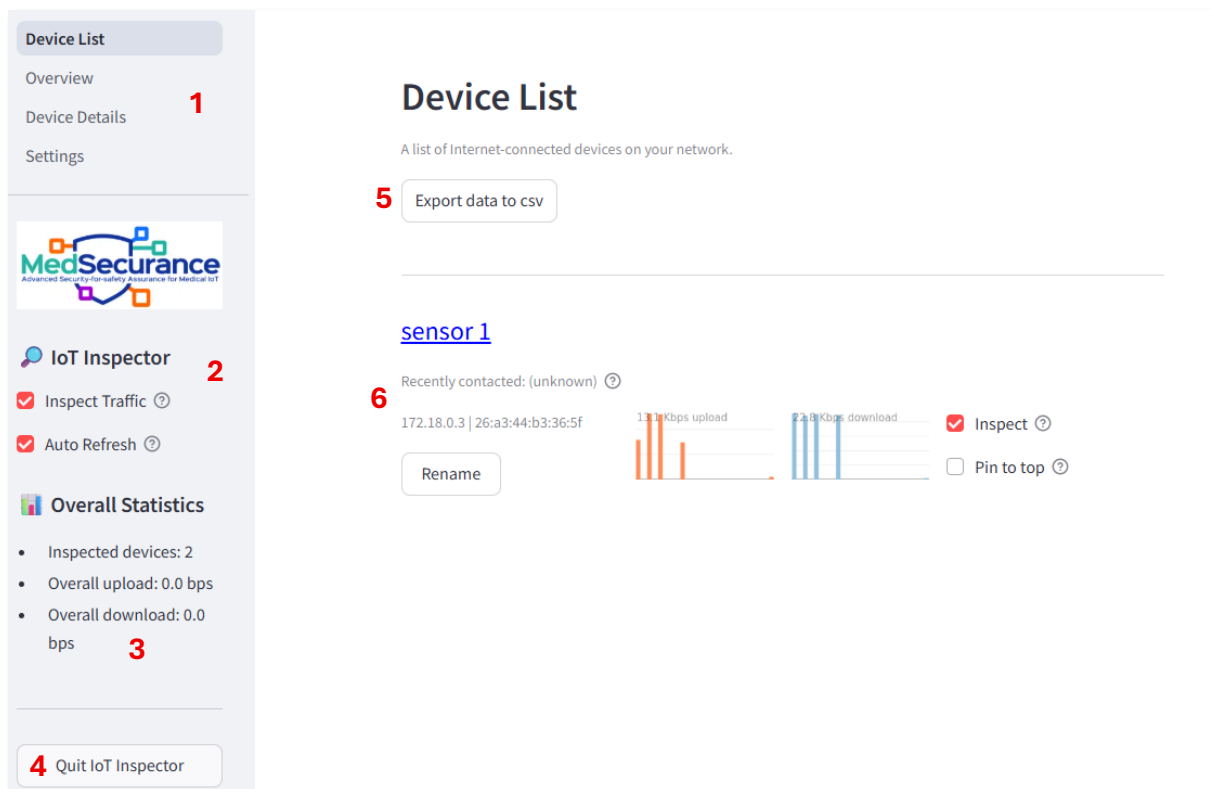


Figure 21: Monitor-Matrix Device List page

3. After accepting the risks consent, the user is presented with the Device List UI as depicted in Figure 21. The following functionality is provided:
 1. The menu area of the UI. More details of each option are provided below
 2. Inspection mode options allow the user to:
 - a) Inspect Traffic: Enables the traffic monitoring of the different devices on the network
 - b) Auto Refresh: Updates the UI results on a fixed time interval
 3. This area provides basic statistics of the Monitor-Matrix module, including the total number of devices detected on the network and the total number of upload and download data from the detected devices.
 4. The button that allows the user to Quit IoT Inspector monitoring service.
 5. Allow the user to export the data depicted in the results area (red number 6) to a csv file.
 6. This is the results area of the Device List UI that provides statistics related to the traffic of the different devices found on the network.

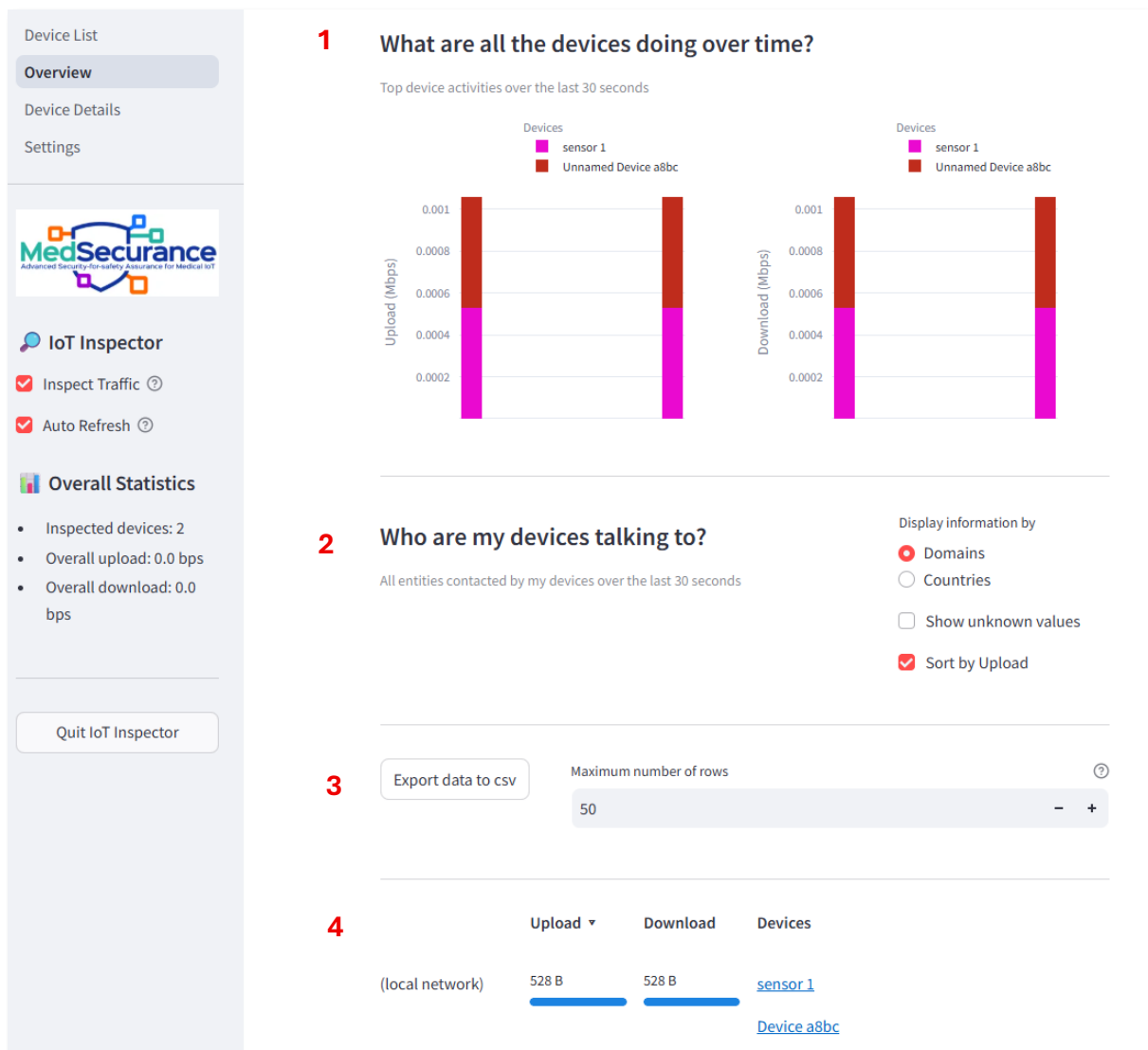


Figure 22: Monitor-Matrix Overview page

- The Overview page is depicted in Figure 22 and provides an overview of the traffic of all devices detected on the network. The different results sections are the following:
 - Box diagrams of the overall traffic in Mbps of all devices on the network.
 - A list of domains and IP addresses that the different devices are communicating with. The user can switch between Domains or Countries using the controls at the left side of the area.
 - The user can export all data as a csv file and also control how many data entries are visible in the UI (For cases where a lot of devices are active on the network)
 - The last area shows the total download and upload data volume of each domain/IP address or country, depending on the option that the users choose to examine.

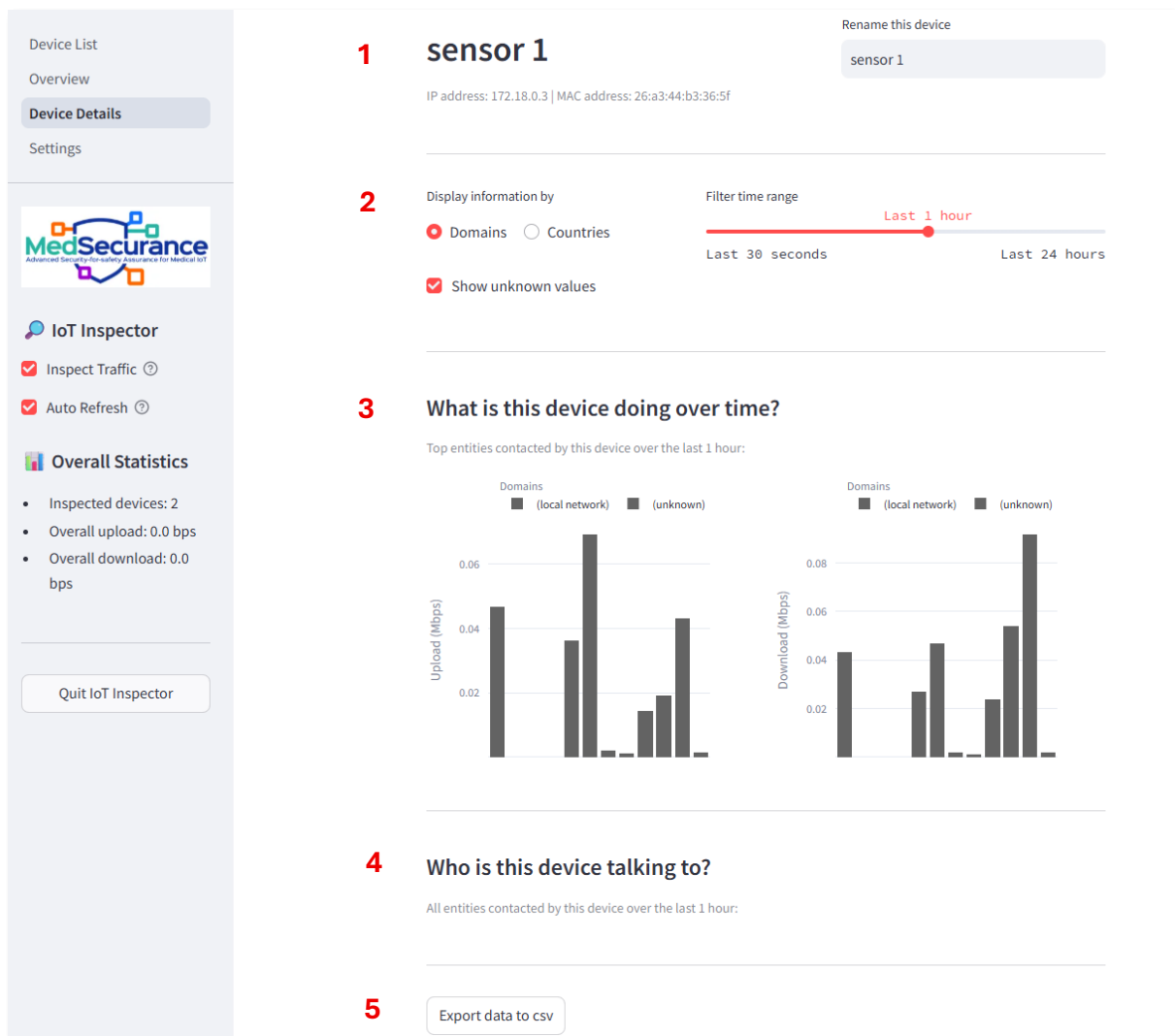


Figure 23: Monitor-Matrix Device Details page

- The Device Details page (Figure 23) allows the user to focus on a single device. Like the other menu options, this section provides the following information:
 - The selected sensor name and basic information
 - The user can switch between Domains or Countries view and define the time window of the report using the slider on the left side of the area. Finally, the “Show unknown values” can be used to show records that the system fail to identify the domain or the IP of the destination.
 - This area shows the traffic (Upload and download in Mbps) of the device to the different domains/IPs.
 - This area shows a list of all domains and IPs that the device is talking to (currently empty in Figure 23).
 - Like the other views, the user can click on “Export data to csv” to download the data as a csv file.

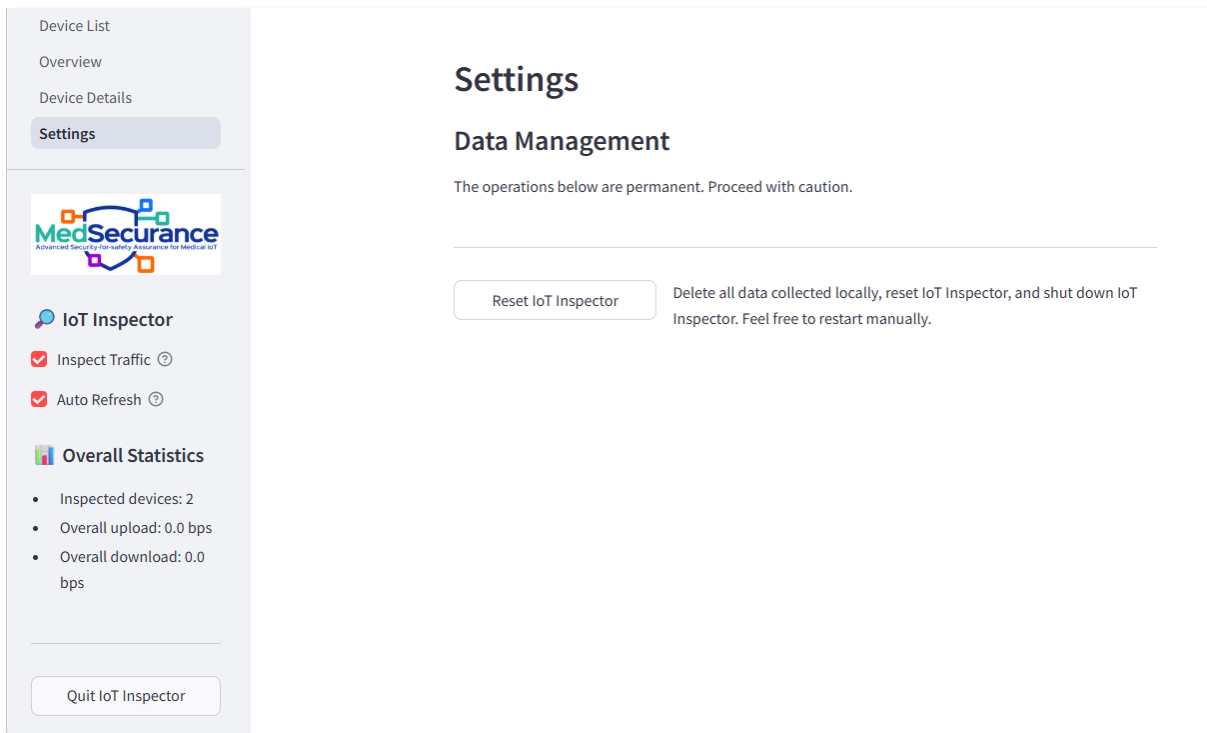


Figure 24: Monitor-Matrix Settings page

- The Settings page is depicted in Figure 24 and allows the user to Reset the Monitor-Matrix instance. That is, re-initialize the Monitor-Matrix to its initial state, equivalent to a fresh install state.

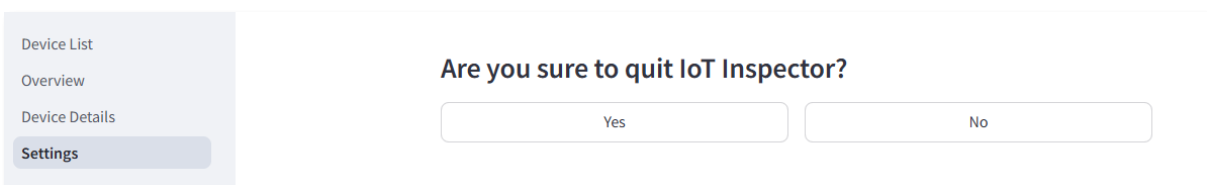


Figure 25: Monitor-Matrix Quit IoT Inspector page

- Finally, following step 3.4 (Quit IoT Inspector), after the user click on the quit button a confirmation page appears as depicted in Figure 25. By clicking “Yes”, the Monitor-Matrix module will be turned off and by clicking “No” the confirmation page will close without termination the module.

Components Functionality Examples

The ICRT is designed to optimize secure, efficient, and compliant communication in IoT/medical environments. Below are detailed examples of how its components (Monitor Matrix, IDS, Device Manager, etc.) operate in real-world scenarios.

1. Monitor Matrix

Purpose: Tracks device communication metrics (destination domain/IP/country, generated traffic) and device behaviour in real time.

Example Scenario:

- **Use Case:** A hospital uses remote patient monitoring devices (e.g., glucose monitors, ECG sensors) that transmit data to a central server.
- **Function:**
 - **Metric Collection:** The Monitor Matrix logs metrics like:
 - **Destination:** Destination IP and Geolocation (critical for real-time alerts).
 - **Generated traffic:** amount of upload and download data between the device and the server (capacity check).
 - **Anomaly Detection:** Detects sudden spikes in traffic (Increased or decreased data transmitted) or unusual communications (e.g., unknown domain/IP or server geo-location).
 - **Unknown device detection:** Check if detected devices already defined in Device Manager.
 - **Output:** Triggers alerts for network administrators to investigate unknown devices attached to the network and unknown/unexpected communication patterns (e.g., unknown personal patient devices, IoMT device communication to unknown serves).

2. Intrusion Detection System (IDS)

Purpose: Identifies and mitigates security threats (e.g., unauthorized access, DDoS attacks).

Example Scenario:

- **Use Case:** A hospital uses IoMT sensors for patient monitoring.
- **Function:**
 - **Signature-Based Detection:** Flags traffic matching known attack patterns (e.g., Mirai botnet signatures).
 - **Behavioural Analysis:** Detects anomalies like:
 - **Unusual Traffic Volume:** 10,000+ requests/second from a single IP (potential DDoS).
 - **Unauthorized Device Access:** A sensor with a spoofed MAC address attempting to join the network.
 - **Response:**
 - Generates email alerts to network administrator.

3. Device Manager

Purpose: Manages device inventory and lifecycle.

Example Scenario:

- **Use Case:** A hospital deploys IoMT-enabled devices for the intensive care units.
- **Function:**
 - **Device Inventory:**
 - Validates devices attached on the network.
 - Monitors unique device IDs and utilized encryption types.
 - Monitoring firmware updates to devices.
 - **Lifecycle Management:**
 - Report outdated devices firmware to prevent vulnerabilities.
 - Report aged devices.

4. Protocol Analyzer

Purpose: Evaluates communication protocols for compliance and performance.

Example Scenario:

- **Use Case:** A healthcare provider uses Bluetooth-enabled wearable devices.
- **Function:**
 - **Protocol Validation:**
 - Checks if devices use **Bluetooth Low Energy (BLE)** with AES-128 encryption.
 - Flags outdated protocols (e.g., BLE 4.0 vs. BLE 5.3).
 - **Compliance Check:**
 - Ensures adherence to **HIPAA** (e.g., data encryption, audit trails).
 - **Recommendations:**
 - Suggests switching to **Wi-Fi 6** for higher throughput if BLE latency is suboptimal.

5. Recommendation Engine

Purpose: Provides actionable recommendations based on analysis.

Example Scenario:

- **Use Case:** A healthcare organization deployed or plan to deploy an IoMT sensors network.
- **Function:**
 - **Contextual Analysis:**
 - Detects mismatched protocols (e.g., HTTP 1.1 vs. MQTT).
 - **Recommendations:**
 - **Protocol Upgrade:** Switch to **MQTT over TLS** for lightweight, secure messaging.
 - **Network Optimization:** Deploy edge gateways to reduce latency.
 - **QoS Settings:** Prioritize meter traffic using **802.1Q VLAN tagging**.